

The classical local Langlands correspondence and its compatibility with Fargues-Scholze

Alexander Bertolini Meri, Linus Hamann

Lecture 1 (ABM) "classical" LLC

F p -adic, G/F quasi-split, $\text{Loett } \mathbb{C}$

$$\mathfrak{L}(G) = \left\{ \varphi : WD \rightarrow {}^L G \right\} / \hat{G}$$

For ss.

$$WD = W_F \rtimes G_a$$

$$\varphi(Fr) \in \hat{G} \times Fr \text{ s.s.}$$

$$S_\varphi := Z_{\hat{G}}(\text{im } \varphi), \quad \overline{S_\varphi} = S_\varphi / Z(\hat{G})^{Fr}$$

Statement. Fix Whittaker datum $(B, \phi) = W$, $\phi : \mathcal{U} \rightarrow \mathbb{C}^\times$

$$\text{Irr}(G) \xleftrightarrow{\text{LLC}_W} \left\{ (\varphi, \rho) : \varphi \in \mathfrak{L}(G), \rho \in \text{Irr}(\pi_0(\overline{S_\varphi})) \right\}$$

$$\pi \longmapsto (\varphi_\pi, \rho_\pi)$$

Properties

1) $\pi_\varphi(G) := \{ \pi : \varphi_\pi = \varphi \}$ is finite.

2) π discrete series $\Leftrightarrow \varphi_\pi$ doesn't factor through a proper Levi of ${}^L G$.

$$\Leftrightarrow |\overline{S_\varphi}| < \infty.$$

3) Compatibility with parabolic induction

$$\varphi^{ss} := \varphi|_{W_F}, \quad \pi \subset \text{IP}^G_P \rho \Rightarrow \varphi_\pi^{ss} = i_M \circ \varphi_\rho^{ss},$$

$$i_M : {}^L M \hookrightarrow {}^L G$$

4) π is W -generic

$$\text{Hom}_{\mathcal{U}(F)}(\pi, \phi) \neq 0$$

$\text{CBFZ} \Rightarrow \varphi_\pi$ has max. possible monodromy

5) π tempered $\Leftrightarrow \text{im}(\varphi_\pi|_{W_F})$ is bounded mod $Z(\hat{G})$.

$\text{DHKMZ} \Rightarrow \varphi_\pi$ smooth in $\tilde{\text{Par}}_G$. $\nearrow \rho_\pi = \text{triv}$

b) ψ tempered $\Rightarrow S\theta_\psi = \sum_{\pi \in \Pi_\psi} \dim \rho_\pi \theta_\pi$ stable dist.

$$\theta_\psi^S = \sum_{\pi \in \Pi_\psi} \text{tr}(\rho_\pi | s) \theta_\pi, \quad s \in S_\psi \text{ lifting some } \bar{s} \in \overline{S_\psi}$$

$\{\theta_\psi^S\}_{\pi_0(\overline{S_\psi})}$ is a basis for $\langle \theta_\pi : \pi \in \Pi_\psi \rangle$

7) Endoscopic character identity (ECI)

(H, s, η) an endoscopic datum for G

$$\eta: {}^L H \hookrightarrow {}^L G, \quad s \in \hat{G}, \quad Z_{\hat{G}}(s)^\circ = \eta(\hat{H})$$

$$\varphi_H \in \mathbb{F}(H), \quad \varphi \in \mathbb{F}(G) \quad \text{s.t.} \quad \eta \circ \varphi_H = \varphi.$$

Th. - Transfer $^H_G S\theta_{\varphi_H} = \theta_\varphi^S$

- f func. G

$$S\theta_{\varphi_H}(f_H) = \theta_\varphi^S(f)$$

f_H func. H

H "smaller" than G

π_{φ_H} smaller than π_φ

$1 \neq \bar{s} \in \pi_0(\overline{S_\psi})$, then $\exists (H, s, \eta)$ nontrivial and φ_H s.t. $\varphi \simeq \eta \circ \varphi_H$.

$\underbrace{\hspace{1cm}}$

(Some of) what is known

GL_n : Harris - Taylor (2001)

Henriart (2000)

Sp_{2n}, SO_{2n+1}

: Arthur (2013)

(Peng) (*) SO_{2n}

Moeglin - Waldspurger

U_n

: Mok (2015)

Extended example parameters of classical gps

$$G = SO_{2n+1}$$

$$\hat{G} = Sp_{2n} \xrightarrow{\text{std}} GL_{2n}$$

$$\varphi \in \Xi(SO_{2n+1}) \rightsquigarrow \text{std} \circ \varphi \in \Xi(GL_{2n})$$

$\text{std} \circ \varphi$ preserves some bilinear form, hence self-dual

$$\text{std} \circ \varphi = \varphi_1 \oplus \dots \oplus \varphi_k.$$

$$= \bigoplus_{j \in J_\varphi} (\varphi_j \oplus \varphi_j^\vee) \oplus \bigoplus_{i \in I_\varphi^+} m_i \varphi_i \oplus \bigoplus_{i \in I_\varphi^-} 2m_i \varphi_i$$

$$\text{sgn}(\varphi_i) : \quad \begin{aligned} f &: M \rightarrow M^\vee \\ f^\vee &= \text{sgn}(\varphi_i) f \end{aligned}$$

$$S_{\varphi, \text{red}} = \prod_{j \in J_\varphi} GL(m_j) \times \prod_{i \in I_\varphi^+} O(m_i) \times \prod_{i \in I_\varphi^-} Sp(2m_i)$$

In particular, φ disc. $\Leftrightarrow J_\varphi = I_\varphi^- = \emptyset$, $m_i = 1$, $\forall i \in I_\varphi^+$

$$|S_\varphi| = |S_{\varphi, \text{red}}| = 2^{|I_\varphi^+|}.$$

Suppose φ is discrete, when is $\pi \in \Pi_\varphi$ supercuspidal?

Th (Mœglin) π as above. π is s.c. $\Leftrightarrow$

- φ_π has no gaps $\sigma \boxtimes n \in \text{Index}(\text{std} \circ \varphi)$, then so does $\sigma \boxtimes n-2$, $n > 2$

- ρ_π is alternating. $\rho_\pi(\sigma \boxtimes 2) = -1$
 $\rho_\pi(\sigma \boxtimes n) = -\rho_\pi(\sigma \boxtimes n-2)$

Inner twists and more

would like to relate classical & CLLC!

Aut. side

reps of each $G_\mathbb{Q}$, $b \in B(G)$

Gal side

sheaves on $BS_\varphi \hookrightarrow \text{Par}_G$

Kottwitz (known U_n , GL_n , SO_{2n} , SO_{2n+1})

KMSW

Chen-Zou Ishimoto

$$\coprod_{b \in B(G)_{\text{basic}}} \text{In}(G_b) \xleftrightarrow{\text{LLCW}} \{(\varphi, \rho) : \varphi \in \Xi(G), \rho \in \text{Irr}(S_\varphi^b)\}$$

$$S_\varphi^b = S_\varphi / [S_\varphi \cap \hat{G}_{\text{der}}]^0$$

$$B(G)_{\text{basic}} \xrightarrow[\sim]{K} \pi_1(G)_\Gamma = X^*(Z(\hat{G})^\Gamma)$$

$$Z(\hat{G})^\Gamma \rightarrow S_\varphi^b$$

$$\text{given } \rho \in \text{In}(S_\varphi^b) \rightsquigarrow \text{central char}(\rho) \in X^*(Z(\hat{G})^\Gamma)$$

Vogan

'pure inner forms'

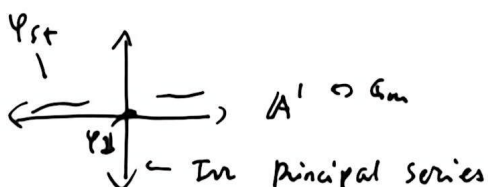
$$\coprod_{\beta \in H^1(F, G)} \text{In}(G_\beta) \hookrightarrow \{ \quad \}$$

$$H^1(F, G) \rightarrow B(G)_{\text{basic}}$$

Assume

Th (BM-01) Kottwitz LLC for G and Levi's, then get

$$\coprod_{b \in B(G)} \text{In}(G_b) \xleftrightarrow{\text{LLCW}} \{(\varphi, \rho) : \varphi \in \Xi(G), \rho \in \text{In}(S_\varphi)\}$$



Passon

Lecture 2 (LH) Compatibility of FS w classical LLC:

- $G/\mathbb{Q}_p$ conn. red. gp (split)
- $\pi(G) = \left\{ \text{smooth irrep of } G(\mathbb{Q}_p) \text{ on } k\text{-vec.sp.} \right\}, k \in \{\bar{\mathbb{Q}}_l, \mathbb{C}\}, \bar{\mathbb{Q}}_l \xrightarrow{\pi} \mathbb{C}.$
 $\downarrow$
discrete top
- $\Phi(G) := \left\{ \psi: W_{\mathbb{Q}_p} \times \mathrm{SL}_2(\bar{\mathbb{Q}}_l) \rightarrow \check{G}(\bar{\mathbb{Q}}_l) \right\} / \check{G}$
 $\downarrow (-)^{ss}$ Frob. s.s.
- $\Phi^{ss}(G) := \left\{ \rho: W_{\mathbb{Q}_p} \rightarrow \check{G}(\bar{\mathbb{Q}}_l) \text{ s.s.} \right\} / \check{G}$

$$g \mapsto \begin{pmatrix} |g|^{1/2} & \\ & |g|^{-1/2} \end{pmatrix}$$

Goal: For LLC_G as in Alexander's talk,

$$\begin{array}{ccc} \pi & \pi(G) & \\ \downarrow & \downarrow \mathrm{LLC}_G & \searrow \mathrm{LLC}_G^{FS} \\ P\pi & \Phi(G) & \xrightarrow{(-)^{ss}} \Phi^{ss}(G) \end{array}$$

Show that this diagram commutes.

Why? Hansen - Mann use this.

Q: $\rho \in \Phi^{ss}(G), \mathrm{LLC}_G^{FS^{-1}}(\rho)$ finite?

$$\mathcal{Z}_G := \mathcal{O}(\mathrm{Par}_G) \sim D(\mathrm{Bun}_G) = D_{\mathrm{cis}, \mathrm{et}}(\mathrm{Bun}_G, \bar{\mathbb{Q}}_l)$$

$\bigcup_{m \in \mathbb{N}} \rightarrow$ semi-simple L-par.

$\varphi \in \Phi^{ss}(G), D(\mathrm{Bun}_G)_\varphi \rightarrow$ allows you to describe this.

What is known?

- A_n (FS)
- C_2 (H.), U_{2n+1}
- $B_n, D_n^{(*)}, U_n$ (Peng)

- There were various restrictions, removed
(Daniels - van Hoften - Kim - Zhang)

$\leadsto$ can handle some central isogenies
 Page 5 $\leadsto$ inner forms H. K. W.

What do we not know?

$G_n, n \geq 2 \dots$? Exceptional groups. (Cotner - Farg, Fu)

Key idea: $\mu \in X_*(T_{\bar{G}_p})^+ \rightarrow \forall \mu \text{ of } \check{G}, \Gamma_\mu: \check{G} \rightarrow GL(V_\mu)$

$\rightarrow T_\mu: D(Bun_G) \rightarrow D(Bun_{\check{G}})^{BW_E}_{\mu = \bar{G}_p}$

$j_{b!}^* T_\mu j_b: D(G_b(\bar{G}_p), \bar{G}_L) \rightarrow D(G(\bar{G}_p), \bar{G}_L)^{BW_E}_\mu$

$b \in B(G, \mu) \rightarrow$ wh. of this space $\xrightarrow{\text{Scholze-Weinstein}} \mu$ is minuscule, p -divisible groups, data of a p -div. gp

$G_b(\bar{G}_p) \xrightarrow{\xi_b} \sum_{\check{G}(\bar{G}_p)} \text{of morphisms } \leq \mu$

$R\Gamma_c(G, b, \mu) \otimes_{H(G_b)} - =: R\Gamma_c(G, b, \mu)[-]$

$A[p^\infty] \xrightarrow{\quad} X_g \mid \bar{\mathbb{F}}_p \text{ to } \mathcal{O}_{\mathbb{F}_p}$

$\uparrow$ $\{ \text{abelian varieties} \}$

$\underline{\Sigma}$ $G = GL_n, \mu = (1, 0, \dots, 0), V_\mu = \text{std}, b = b_{1/n}$

$R\Gamma_c(G, b, \mu)[-] := R\Gamma_c(LT_{\check{G}}, \bar{G}_L[d](-\frac{d}{2}))[-]$

$\xrightarrow{\text{Lubin-Tate tower}}$

$G_b = GL_1(D_{1/n})$

Th. (Harris - Taylor) $\pi \in \Pi(GL_n)$ s.c. rep'n. $JL^{-1}(\pi) = \rho \in \Pi(D_{1/n})$

$R\Gamma_c(G, b, \mu)[\rho] \simeq \pi \boxtimes \rho \pi$

$\mu \in VB(\check{G})^{BW_E}_\mu \rightarrow \Gamma_\mu \circ \rho^{\text{univ}}$

$j_{b!}^* T_\mu j_b! \rho \xrightarrow{\quad} \check{G}_L$

$\varphi_\rho^{FS} \simeq \varphi_\pi^{FS}$

$\Gamma_\mu \circ \varphi_\rho^{FS} \parallel \varphi_\rho^{FS}$

The properties Alexander mentioned come from studying

$$L^2_{\text{disc}}(G(A) \setminus G(\mathbb{A})) \quad G/A, \quad G_{\text{ap}} \simeq G$$

$$\hat{\pi} \simeq \bigotimes_{v \leq \infty} \pi_v$$

$$-(G, X)^{\mu \in X_k(T)^+} \quad h: \text{Res}_{\mathbb{C}|\mathbb{R}} G_M \rightarrow G_{\mathbb{R}} \\ \hookrightarrow \\ G_{\mathbb{R}} \quad \quad \quad K \subset G(A_f)$$

$$G(G) \times G(A_f) / K$$

E reflex field

$$\sim \text{Sh}(G, X)_K \xrightarrow[\text{JK}]{G(A_f)} \text{Sh}(G, X)_K^*$$

$$R\Gamma_{!x}(G, X)_K := R\Gamma(\mathrm{Sh}(G, X)^*_K, j_{K!x}(\widehat{\mathcal{O}}_K[d](\frac{d}{2})))$$

$$R\Gamma_{!,*}(\mathcal{L}, X) = \text{colim}_K \dots$$

$$\bar{G}_E [G(A_f) \times_{\text{Gal}(\bar{E}|E)}]$$

$$R\Gamma_{\mathcal{Y}}(G, X, \mathcal{I}) \simeq \bigoplus_{\pi \in \text{Ld}_{\mathcal{Y}}} \pi^* \mathcal{M}_F \otimes_{A \pi, \mathcal{I}}^{\mathbb{L}} \mathcal{M}_F$$

(can find globalisation $\pi \in L^2_{\text{resp}}$ s.t.

$$A\pi_{\infty,3} \simeq \Gamma_{\mu} \circ \rho_{\pi}^{\oplus n} \rightarrow \text{global Galois rep.}$$

$$\bar{a}_p \simeq c, \quad E_p | a_p \quad . \quad r_\mu \cdot \varphi_\pi |_{u \in p} \simeq r_\mu \cdot \underbrace{\varphi_{p_r}}_{ss}$$

T6. $\exists$ a sheaf $F_{1,x} \in D(\mathcal{B}ar_n)$ i.e. $j_{1*}^y T_M F_{1,x}$

$$\Gamma_\mu \circ \varphi_{\pi_P}^{SS} \simeq \Gamma_\mu \circ \varphi_{\pi_P}^{FS} \simeq \begin{matrix} R\Gamma_{!x}(A, X) \\ \downarrow \\ \mathcal{L}(C_P) \times W_{E_P} \end{matrix}$$

For sufficiently nice minuscule μ , $\Rightarrow \varphi_{\pi_p}^{\text{JS}} = \varphi_{\pi_p}^{\text{PS}}$

Subtle but interesting point,

Want π_P s.c. rep.

to assume

Better: π_P sits in a packet w/ only s.c. rep.

$\Leftrightarrow \mathcal{P}\pi_P$ is supercuspidal

Can actually reduce to this case!

Ex. $G = \mathrm{GSp}_4$ $G, \mu = (1, 1, 0, 0)$ $\check{G} = \mathrm{GSpin}_5 \simeq \mathrm{GSp}_4$

$\left. \begin{array}{l} \text{Spin}_2 \\ \text{std} \end{array} \right\} \mathrm{GL}_4$

$b \in B(G, \mu) \wedge B(G)_{\text{basic}}$

$G_b = \mathrm{GU}_2(D)$

$\text{std. } \rho \simeq \rho_0 \oplus \chi \boxtimes \nu(2)$
 $\hookrightarrow 2\text{-dim rep of } \mathrm{SL}(2)$

$\overline{S}_\psi = \mathbb{Z}/2\mathbb{Z}$

$\pi_P(\mathrm{GSp}_4) = \{ \pi_{sc}, \pi_{nsc} \}$
 $\nwarrow$ end. char identity
 $\pi_P(\mathrm{GU}_2(D)) = \{ p_{sc}, p_{nsc} \}$

$\left(\mathrm{R}\Gamma_c(G, b, \mu) [p_{nsc}] \right)_{sc} \in \mathrm{K}_0(G(\mathbb{Q}_p))_{sc}$

|| HKW, Mieda
 $\sim 2\pi_{sc}$

$\varphi_{p_{nsc}}^{\mathrm{FS}} = \varphi_{\pi_{sc}}^{\mathrm{FS}}$

|| induction ||

$\varphi_{p_{nsc}}^{\mathrm{ss}} = p_{\pi_{sc}}^{\mathrm{ss}}$

done